

Arithmetic-Geometric Structure of Agrawal’s Conjecture and the Torus Dimension Count Behind the Double Condition

(AG track working document — v0.1, item AG2)

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Abstract

We reframe Agrawal’s conjecture and the Lenstra–Pomerance counterexample mechanism in the language of one-dimensional algebraic groups. The main new result (item AG2 of `doc/arith-geom-plan.md`) upgrades the empirical finding A11 of this project (`data/FINDINGS.md`) to a structural statement: for *every* AKS modulus r , the constructive (Lenstra–Pomerance-style) route forces smoothness control on the $\mathbb{F}_p$ -point orders of a product of cyclotomic tori of total dimension at least 2, and the minimum 2 is attained exactly by the split and nonsplit one-dimensional tori — orders $p - 1$ and $p + 1$ — i.e. by the Carmichael + Lucas–Carmichael double condition (equivalently, twin-smoothness). The key identity is an inversion collapse: when $d = \text{ord}_r(p)$ is even, the unique involution of $(\mathbb{Z}/r)^\times$ is -1 , forcing $(X - 1)^{p^{d/2}-1} \equiv -X^{-1} \pmod{p, \Phi_r(X)}$ and collapsing the order of $X - 1$ into the cyclotomic levels $e \mid d/2$. Since $\sum_{e \mid m} \varphi(e) = m$, the burden is $d/2$ for even d and d for odd d ; validity of a counterexample forces $d \geq 3$, so the burden is minimized at $d = 4$ (possible iff $4 \mid r - 1$), value 2, never 1. All identities and the resulting burden table are verified numerically (`code/ag2_torus_check.py`), reproducing A11’s measured table for all eleven moduli tested there.

1 The Frobenius-lift dictionary

This section fixes the geometric reading used throughout; full exposition is item AG1 and will accrete here. Write $T(-1, n, r)$ for the AKS congruence

$$(X - 1)^n \equiv X^n - 1 \pmod{n, X^r - 1}, \tag{1}$$

r an odd prime not dividing n .

Proposition 1.1 (primality as a Frobenius lift). *Let $A_n = (\mathbb{Z}/n)[X]/(X^r - 1)$, the mod- n group algebra of μ_r , and let $\sigma_n : X \mapsto X^n$ be the n -th Adams operation (well defined since $X^r = 1$). If n is prime, then the n -power map on A_n coincides with σ_n ; that is, $f(X)^n \equiv f(X^n) \pmod{n, X^r - 1}$ for every f . The congruence (1) is this coincidence tested on the single element $f = X - 1$.*

Proof. For n prime the n -power map is a ring endomorphism (freshman’s dream) and fixes $\mathbb{Z}/n$ (Fermat), so $f(X)^n \equiv f^{(n)}(X^n) \equiv f(X^n)$, where $f^{(n)}$ raises coefficients to the n -th power. $\square$

In λ -ring language: $\mathbb{Z}[X]/(X^r - 1) = \mathbb{Z}[\mathbb{Z}/r]$ carries the Adams operations $\psi^k(X) = X^k$, and “ n is prime” is exactly “ ψ^n is a Frobenius lift at n ” (Wilkerson’s criterion direction); (1) is the lift condition at one section of μ_r . **[TODO: AG1: cite Wilkerson, Borger, and the pseudofield formalism of Lenstra–Pomerance.]**

Lemma 1.2 (the exceptional clause). $n^2 \equiv 1 \pmod{r}$ iff σ_n , viewed as the element $n \bmod r$ of $(\mathbb{Z}/r)^\times \cong \text{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q})$, lies in $\text{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+) = \{1, \text{complex conjugation}\}$.

Proof. $(\mathbb{Z}/r)^\times$ is cyclic of even order $r-1$; its unique element of order 2 is -1 , which acts on μ_r as inversion, i.e. as complex conjugation, the generator of $\text{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+)$. $\square$

Agrawal's conjecture thus asserts: *a composite Frobenius impostor on the single section $X-1$ must act trivially on the maximal real subfield.* A valid counterexample is a composite n with $T(-1, n, r)$ and $t := \text{ord}_r(n) > 2$.

2 Setup: cyclotomic tori and the mechanism

2.1 Cyclotomic tori

For $e \geq 1$ let Φ_e denote the e -th cyclotomic polynomial, so that

$$p^d - 1 = \prod_{e|d} \Phi_e(p). \quad (2)$$

Over $\mathbb{F}_p$, the Weil restriction $\text{Res}_{\mathbb{F}_{p^d}/\mathbb{F}_p} \mathbb{G}_m$ decomposes up to isogeny as $\prod_{e|d} T_e$, where T_e is the cyclotomic torus of dimension $\varphi(e)$ split by $\mathbb{F}_{p^e}$, with $\#T_e(\mathbb{F}_p) = \Phi_e(p)$ [7, 8]. The two *one-dimensional* tori are $T_1 = \mathbb{G}_m$ ($\#T_1(\mathbb{F}_p) = p-1$, the split torus) and T_2 the norm-one torus of $\mathbb{F}_{p^2}/\mathbb{F}_p$ ($\#T_2(\mathbb{F}_p) = p+1$, the nonsplit twist); $\varphi(e) = 1$ iff $e \in \{1, 2\}$. We refer to the index e in (2) as the *level*.

2.2 The order of $X-1$

Fix primes $p \neq r$, $p \nmid n$, and let $d = d_p := \text{ord}_r(p)$. Then

$$\mathbb{F}_p[X]/(X^r - 1) \cong \mathbb{F}_p \times \mathbb{F}_p[X]/(\Phi_r(X)), \quad \mathbb{F}_p[X]/(\Phi_r(X)) \cong \prod_{i=1}^{(r-1)/d} \mathbb{F}_{p^d},$$

the factors indexed by the orbits of $\langle p \rangle$ on $(\mathbb{Z}/r)^\times$; in the i -th factor X maps to a primitive r -th root of unity ζ_i . The element $X-1$ vanishes in the $\mathbb{F}_p$ factor (where (1) holds trivially: both sides are 0) and is a unit in $\mathbb{F}_p[X]/\Phi_r$ since $\Phi_r(1) = r \neq 0$ in $\mathbb{F}_p$.

Definition 2.1. $\rho_r(p)$ is the multiplicative order of the image of $X-1$ in $(\mathbb{F}_p[X]/\Phi_r(X))^\times$, i.e. $\rho_r(p) = \text{lcm}_i \text{ord}(\zeta_i - 1)$.

For $r = 5$ this is the quantity $\rho(p)$ of Vána [6], with $\rho(p) \mid 10(p^2 - 1)$; Corollary 3.3 below recovers this.

2.3 The Lenstra–Pomerance mechanism

Definition 2.2 (L–P mechanism). A squarefree composite $n = p_1 \cdots p_k$ coprime to r is produced by the L–P mechanism if for every i there is $j_i \geq 0$ with

$$n \equiv p_i^{j_i} \pmod{\text{lcm}(r, \rho_r(p_i))}, \quad (3)$$

and $\text{ord}_r(n) > 2$ (validity).

The original $r = 5$ construction [2, 4] is the instance $j_i = 1$: the conditions $p \equiv 3 \pmod{80}$, $p - 1 \mid n - 1$, $p + 1 \mid n + 1$ give $n \equiv p \pmod{\text{lcm}(5, \rho_5(p))}$ once $\rho_5(p) \mid 10(p^2 - 1)$ is known. Our entire Track A pipeline (PLAN.md) is a search inside Definition 2.2.

Remark 2.3 (scope). The mechanism is sufficient, not necessary, for (1): the component equation $(\zeta - 1)^n = \zeta^n - 1$ could conceivably hold “accidentally” for some p , outside the Frobenius-matching family (3). No such solutions are known, and no way to construct them is known; every published counterexample construction [2, 4, 5] is an instance of Definition 2.2. All results below quantify *this mechanism*, in line with the scope discipline of the negative-result paper (doc/negative-result-plan.md).

3 The torus dimension count

3.1 Frobenius matching and the inversion collapse

Lemma 3.1 (Frobenius matching). *With notation as above:*

- (i) $(X - 1)^{p^k} \equiv X^{p^k} - 1 \pmod{p, X^r - 1}$ for all $k \geq 0$.
- (ii) If $n \equiv p^j \pmod{\text{lcm}(r, \rho_r(p))}$ then $T(-1, n, r)$ holds modulo p .
- (iii) Conversely, (3) forces $n \equiv p^j \pmod{r}$, hence $\text{ord}_r(n) \mid d_p$. In particular a valid counterexample ($\text{ord}_r(n) > 2$) forces $d_p \geq 3$ for every $p \mid n$.

Proof. (i) is the iterated Frobenius endomorphism of $\mathbb{F}_p$ -algebras. (ii): write $n = p^j + c \cdot \text{lcm}(r, \rho_r(p))$. In the $\mathbb{F}_p$ factor both sides of (1) vanish. In $\mathbb{F}_p[X]/\Phi_r$: $(X - 1)^n = (X - 1)^{p^j} ((X - 1)^{\rho_r(p)})^{c'} = (X - 1)^{p^j} \equiv X^{p^j} - 1$ by (i), and $X^{p^j} = X^{n \bmod r} = X^n$ since $X^r \equiv 1 \pmod{\Phi_r}$ (as $X^r - 1 = (X - 1)\Phi_r(X)$ for r prime). (iii): reduce (3) modulo r ; then $n \in \langle p \rangle \subseteq (\mathbb{Z}/r)^\times$, so $\text{ord}_r(n)$ divides $\#\langle p \rangle = d_p$, and $\text{ord}_r(n) \geq 3$ gives $d_p \geq 3$. $\square$

Lemma 3.2 (inversion collapse). *Suppose $d = \text{ord}_r(p)$ is even. Then, in $\mathbb{F}_p[X]/\Phi_r(X)$,*

$$(X - 1)^{p^{d/2} - 1} = -X^{-1},$$

and consequently

$$\rho_r(p) \mid 2r(p^{d/2} - 1) = 2r \prod_{e \mid d/2} \Phi_e(p). \quad (4)$$

Proof. $(\mathbb{Z}/r)^\times$ is cyclic, so its unique element of order 2 is -1 ; since $\text{ord}_r(p) = d$, the element $p^{d/2}$ has order 2, whence $p^{d/2} \equiv -1 \pmod{r}$. In each field component, with $X \mapsto \zeta$ a primitive r -th root of unity, iterated Frobenius gives

$$(\zeta - 1)^{p^{d/2}} = \zeta^{p^{d/2}} - 1 = \zeta^{-1} - 1 = -\zeta^{-1}(\zeta - 1),$$

and dividing by $\zeta - 1 \neq 0$ yields $(\zeta - 1)^{p^{d/2} - 1} = -\zeta^{-1}$. This holds in every component, hence in the product ring. Finally $(-X^{-1})^{2r} = X^{-2r} = 1$, so $(X - 1)^{2r(p^{d/2} - 1)} = 1$, giving (4); the product form is (2) applied to $p^{d/2} - 1$. $\square$

Corollary 3.3 (Vána’s bound). *For $r = 5$ and $p \equiv \pm 2 \pmod{5}$ (so $d = 4$), $\rho_5(p) \mid 10(p^2 - 1)$.*

Definition 3.4 (support level). *Set*

$$d' = d'(p) := \begin{cases} d/2, & d \text{ even}, \\ d, & d \text{ odd}. \end{cases}$$

For odd d the trivial bound $\rho_r(p) \mid p^d - 1 = \prod_{e \mid d} \Phi_e(p)$ applies (orders in a product of copies of $\mathbb{F}_{p^d}^\times$), so in both cases

$$\rho_r(p) \mid 2r \prod_{e \mid d'} \Phi_e(p) : \quad (5)$$

the p -dependent part of $\rho_r(p)$ is supported on the cyclotomic levels $e \mid d'$.

3.2 Generic sharpness

The bound (5) is an upper bound on support; the mechanism's cost analysis also needs it to be generically *attained* — otherwise a harvest could fish for primes p where $\rho_r(p)$ collapses further.

Proposition 3.5 (generic sharpness; heuristic + empirical). *Let $\ell > 2rd$ be a prime with $\ell \mid \Phi_e(p)$ for some $e \mid d'$. Then $\ell \mid \rho_r(p)$ unless in every component the image of $X - 1$ is an ℓ -th power in $\mathbb{F}_{p^d}^\times$. Modelling the components as independent uniform elements, the exceptional probability is $\ell^{-(r-1)/d} \leq 1/\ell$. Consequently, for all p outside a set of (heuristic) density $O(\sum_\ell 1/\ell)$, the large-prime part of $\prod_{e \mid d'} \Phi_e(p)$ divides $\rho_r(p)$, and requiring “ $\rho_r(p)$ is B -smooth” is equivalent to requiring “ $\Phi_e(p)$ is B -smooth for every $e \mid d'$ ”.*

Evidence. $\ell \nmid \rho_r(p)$ iff the image of $X - 1$ has trivial ℓ -part in each cyclic component group $\mathbb{F}_{p^d}^\times$, i.e. is an ℓ -th power in every component whose order ℓ divides; the ℓ -th powers have index ℓ in each component containing ℓ -torsion. Rigorously controlling the image of $\zeta - 1$ is an open Artin-type problem; empirically (`code/ag2_torus_check.py`, 67 sample primes across $r \in \{5, 7, 11, 13, 19, 23, 41\}$, $d \in \{3, 4, 5, 6, 11\}$): all 37 tested large primes ℓ divided $\rho_r(p)$ (0 exceptions; ≈ 0.19 expected under the $1/\ell$ model), and in all 67 cases removing the top level $e = d'$ from the bound (5) broke it (the top level is genuinely present in $\rho_r(p)$). $\square$

3.3 The dimension-count theorem

Theorem 3.6 (torus dimension count). *Let r be an odd prime and let n be produced by the L - P mechanism (Definition 2.2) with $\text{ord}_r(n) > 2$. For each $p \mid n$ put $d_p = \text{ord}_r(p)$ and d'_p as in Definition 3.4. Assume generic sharpness (Proposition 3.5) at p . Then, for the CRT system (3) to admit assembly over many primes with a common B -smooth modulus (the setting of the Track-A pipeline), each $p \mid n$ must satisfy:*

$$\Phi_e(p) \text{ is } B\text{-smooth for every } e \mid d'_p \iff \# \left(\prod_{e \mid d'_p} T_e \right) (\mathbb{F}_p) \text{ is } B\text{-smooth},$$

a smoothness condition on the $\mathbb{F}_p$ -points of a torus of dimension

$$\dim \prod_{e \mid d'_p} T_e = \sum_{e \mid d'_p} \varphi(e) = d'_p = \begin{cases} d_p/2, & d_p \text{ even}, \\ d_p, & d_p \text{ odd}. \end{cases}$$

Moreover:

- (a) $d'_p \geq 2$ for every $p \mid n$: the burden is never a single one-dimensional torus. (Indeed $d'_p = 1$ would force $d_p \in \{1, 2\}$, contradicting $d_p \geq 3$ from Lemma 3.1(iii).)
- (b) $d'_p = 2$ iff $d_p = 4$, in which case the level set is $\{1, 2\}$ and the condition is: $p-1 = \#T_1(\mathbb{F}_p)$ and $p+1 = \#T_2(\mathbb{F}_p)$ both B -smooth — the Carmichael + Lucas–Carmichael double condition, i.e. twin-smoothness of $(\frac{p-1}{2}, \frac{p+1}{2})$.
- (c) $d_p = 4$ is realizable iff $4 \mid r-1$; hence for $r \equiv 1 \pmod{4}$ the minimal burden is 2, identical in structure to the $r = 5$ case, while for $r \equiv 3 \pmod{4}$ every admissible d_p gives burden ≥ 3 .
- (d) The minimal burden as a function of r is

$$\text{burden}(r) = \min_{\substack{d \mid r-1 \\ d \geq 3}} \begin{cases} d/2, & d \text{ even} \\ d, & d \text{ odd} \end{cases}$$

which reproduces the measured A11 table for all eleven moduli tested there (Section 4).

Proof. The support statement is (5) plus Proposition 3.5; the dimension computation is $\sum_{e \mid m} \varphi(e) = m$ with $m = d'_p$, together with $\dim T_e = \varphi(e)$ and $\#T_e(\mathbb{F}_p) = \Phi_e(p)$ (Section 2). (a): $d_p \geq 3$ by Lemma 3.1(iii); $d'_p = 1$ requires $d_p \leq 2$. (b): $d'_p = 2$ with d_p even means $d_p = 4$ and levels $\{e : e \mid 2\} = \{1, 2\}$; $d'_p = 2$ with d_p odd is impossible ($d_p = 2$ is even). With p odd, B -smoothness of $p \mp 1 = 2m, 2(m+1)$ is twin-smoothness of the pair $(m, m+1)$ up to the factor 2. (c): $d_p = 4$ requires $4 \mid \#(\mathbb{Z}/r)^\times = r-1$; conversely if $4 \mid r-1$ the cyclic group $(\mathbb{Z}/r)^\times$ has elements of order 4, and primes p in those residue classes exist (Dirichlet), with $\text{ord}_r(n) = 4 \mid d_p$ satisfiable. For $r \equiv 3 \pmod{4}$: even $d \mid r-1$ satisfy $d \equiv 2 \pmod{4}$, so even $d \geq 3$ means $d \geq 6$, burden $d/2 \geq 3$; odd $d \geq 3$ gives burden $d \geq 3$. (d) is the minimization of d'_p over admissible d_p . $\square$

Corollary 3.7 (structural closure of option (c)). *For every AKS modulus r , the L–P mechanism requires each pool prime to satisfy a smoothness condition on the point-orders of a torus of dimension at least 2; no modulus reduces the requirement to a single one-dimensional torus (a single condition of the shape “ $p \pm 1$ smooth”). The minimum, dimension exactly 2, is achieved precisely by the split/nonsplit pair $\{T_1, T_2\}$ — the twin-smooth double condition — and only for $r \equiv 1 \pmod{4}$. The double condition measured empirically in A11 is therefore intrinsic: it is the unique minimizer of a dimension count, not an artifact of $r = 5$.*

Remark 3.8 (cost interpretation). Under the standard Dickman heuristic, B -smoothness of $\Phi_e(p)$, a quantity of size $p^{\varphi(e)}$, has cost governed by $u = \varphi(e) \log p / \log B$; total burden across the level set is governed by $\sum_{e \mid d'} \varphi(e) = d'$. Burden 2 is the twin-smooth regime this project has harvested at scale; burden ≥ 3 requires smooth values of $\Phi_e(p) \sim p^{\varphi(e)}$ with $\varphi(e) \geq 2$, the same quantities whose *non*-smoothness underpins torus-based cryptography [8]. This is the structural content of A11’s “every other r is strictly worse”.

Remark 3.9 (what remains conditional). Lemmas 3.1 and 3.2, the support bound (5), and parts (a)–(d) of Theorem 3.6 as statements about the support of $\rho_r(p)$ are unconditional. The conditional ingredient is generic sharpness (Proposition 3.5): a harvest could in principle target the thin set of primes where $\rho_r(p)$ collapses below (5). Such primes have heuristic density $O(1/\ell)$ per large prime ℓ avoided — i.e. the “collapsed” pool is far thinner than the twin-smooth pool itself — so this escape is uneconomical, but it is recorded here as the localized gap per gate AG-G1 (doc/arith-geom-plan.md).

4 Numerical validation

code/ag2_torus_check.py (deterministic; output in data/generated/ag2_torus_check.json) verifies, over 67 sample primes spanning $(r, d) \in \{(5, 4), (13, 4), (41, 4), (7, 3), (7, 6), (19, 3), (11, 5), (23, 11)\}$:

check	statement	result
V1	Frobenius matching (Lemma 3.1(i))	67/67
V2	collapse identity $(X - 1)^{p^{d/2}-1} = -X^{-1}$ (Lemma 3.2)	36/36
V3	exact $\rho_r(p)$ computed within bound (5)	67/67
V4	large primes of levels $e \mid d'$ divide $\rho_r(p)$ (Prop. 3.5)	37/37
V5	top level $e = d'$ genuinely needed in $\rho_r(p)$	67/67
V6	burden formula (Thm. 3.6(d)) vs. A11 table	11/11

V4's 0 exceptions are consistent with the $1/\ell$ model (≈ 0.19 misses expected over this sample). V6 reproduces, for $r \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$, both the burden value and the minimizing d of the A11 measurement (data/FINDINGS.md, 2026-08-24).

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